$$\int_U (au+bv)D^\alpha\varphi = (-1)^k\int_U (aD^\alpha u + bD^\alpha v)\varphi$$

Thm: If $u, v \in W^{k,p}(U)$, then for any α with $|\alpha| \le k$,

$$D^\alpha(au+bv) = aD^\alpha u + bD^\alpha v \quad \forall a, b \in \mathbb{R}.$$

Pf: For any $\varphi \in C_c^\infty(U)$,

$$\int_U (au+bv)D^\alpha\varphi = a\int_U uD^\alpha\varphi + b\int_U vD^\alpha\varphi$$

$$= a(-1)^{|\alpha|}\int_U \varphi D^\alpha u + b(-1)^{|\alpha|}\int_U \varphi D^\alpha v$$

$v = D^\alpha u \Rightarrow$

$$\int_U uD^\alpha\varphi = (-1)^{|\alpha|}\int_U v\varphi \quad \forall \varphi \in C_c^\infty(U)$$

$$= (-1)^{|\alpha|} \int_U (a D^\alpha u + b D^\alpha v)\varphi. \quad \#$$

<u>Thm:</u> $\|\cdot\|_{W^{k,p}(U)}$ is a norm on $W^{k,p}(U)$.

<u>Pf:</u> 1). Clearly, $\|u\|_{W^{k,p}(U)} \geq 0$ for all $u \in W^{k,p}(U)$ and $\|0\|_{W^{k,p}(U)} = 0$.

$$\|u\|_{W^{k,p}(U)} = \begin{cases} \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|^p_{L^p(U)} \right)^{1/p}, & 1 \leq p < \infty \\ \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^\infty(U)}, & p = \infty \end{cases}$$

includes $\alpha = (0, \dots, 0)$ corresponding to $\|u\|_{L^p(U)}$

includes $\|u\|_{L^\infty(U)}$

Suppose $\|u\|_{W^{k,p}(U)} = 0$. Then $\|u\|_{L^p(U)} = 0$ and hence $u = 0$ a.e.,

so $u = 0$ in $W^{k,p}(U)$.

2). $\|au\|_{W^{k,p}(U)} = |a| \, \|u\|_{W^{k,p}(U)}$ for all $a \in \mathbb{R}$ since

$D^\alpha(au) = a D^\alpha u$ and the $L^p(U)$ norm has this property.

3). Let $u, v \in W^{k,p}(U)$. If $1 \leq p < \infty$,

$$\|u+v\|_{W^{k,p}(U)} = \left(\sum_{|\alpha| \leq k} \|\underbrace{D^\alpha(u+v)}_{D^\alpha u + D^\alpha v}\|^p_{L^p(U)} \right)^{1/p}$$

$\|u+v\|_{W^{k,p}(U)} \leq \|u\|_{W^{k,p}(U)} + \|v\|_{W^{k,p}(U)}$

$$\leq \left[\sum_{|\alpha| \leq k} \left(\|D^\alpha u\|_{L^p(U)} + \|D^\alpha v\|_{L^p(U)} \right)^p \right]^{1/p}$$

$$\leq \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|^p_{L^p(U)} \right)^{1/p} + \left(\sum_{|\alpha| \leq k} \|D^\alpha v\|^p_{L^p(U)} \right)^{1/p}$$

$$= \|u\|_{W^{k,p}(U)} + \|v\|_{W^{k,p}(U)}.$$

<u>HW:</u> Give the proof of (3) for $p = \infty$. #

$$\ell^p = \left\{ a = (a_1, a_2, \dots) : \|a\|_{\ell^p} = \left(\sum_{n=1}^{\infty} |a_n|^p \right)^{1/p} \right\}$$

$$\|a+b\|_{\ell^p} \leq \|a\|_{\ell^p} + \|b\|_{\ell^p}$$

Thm: $W^{k,p}(U)$ is a Banach space.

Pf: Let (u_j) be a Cauchy seq in $W^{k,p}(U)$, i.e.,

$$\|u_j - u_\ell\|_{W^{k,p}(U)} \to 0 \quad \text{as } j,\ell \to \infty.$$

Since

$$\|u_j - u_\ell\|_{W^{k,p}(U)} = \begin{cases} \left(\sum_{|\alpha| \le k} \|D^\alpha u_j - D^\alpha u_\ell\|^p_{L^p(U)} \right)^{1/p}, & 1 \le p < \infty \\ \sum_{|\alpha| \le k} \|D^\alpha u_j - D^\alpha u_\ell\|_{L^\infty(U)}, & p = \infty, \end{cases}$$

then for each α with $|\alpha| \leq k$,

$$\| D^\alpha u_j - D^\alpha u_\ell \|_{L^p(U)} \to 0 \quad \text{as } j, \ell \to \infty,$$

i.e., $(D^\alpha u_j)$ is a Cauchy seq in $L^p(U)$. Since $L^p(U)$ is complet, then $\exists\, u_\alpha \in L^p(U)$ st

$$\| D^\alpha u_j - u_\alpha \|_{L^p(U)} \to 0 \quad \text{as } j \to \infty.$$

For $\alpha = (0, \dots, 0)$, this gives a function $u = u_{(0,\dots,0)} \in L^p(U)$ st

$$\| u_j \to u \|_{L^p(U)} \to 0 \quad \text{as } j \to u.$$

We will show that $u_\alpha = D^\alpha u$. We have

$$\left| \int_U u_j D^\alpha \varphi - \int_U u D^\alpha \varphi \right|$$

$$= \left| \int_U (u_j - u) D^\alpha \varphi \right|$$

$$\leq \|u_j - u\|_{L^p(U)} \|D^\alpha \varphi\|_{L^q(U)} \to 0$$

<u>To show:</u> $\exists u \in W^{k,p}(U)$ st

$$\|u_j - u\|_{W^{k,p}(U)} \to 0$$

$$\|u_j - u\|_{W^{k,p}(U)} = \begin{cases} \left(\sum_{|\alpha| \leq k} \|D^\alpha u_j - D^\alpha u\|_{L^p(U)} \right)^{1/p} & 1 \leq p < \infty \\ \sum_{|\alpha| \leq k} \|D^\alpha u_j - D^\alpha u\|_{L^\infty(U)} & p = \infty \end{cases}$$

by the Hölder inequality, where $\frac{1}{p}+\frac{1}{q}=1$. Similarly, $\int_U \varphi D^\alpha u_j \longrightarrow \int \varphi u_\alpha$. So passing to the limit in

$$\int_U u_j D^\alpha \varphi = (-1)^{|\alpha|} \int_U \varphi D^\alpha u_j$$

gives

$$\int_U u D^\alpha \varphi = (-1)^{|\alpha|} \int_U \varphi u_\alpha .$$

Since this holds for all $\varphi \in C_c^\infty(U)$, $u_\alpha = D^\alpha u$ and hence $u \in W^{k,p}(U)$. Then

$$\|u_j - u\|_{W^{k,p}(U)} = \begin{cases} \left(\sum_{|\alpha| \le k} \|D^\alpha u_j - u_\alpha\|^p_{L^p(U)} \right)^{1/p}, & 1 \le p < \infty \\ \sum_{|\alpha| \le k} \|D^\alpha u_j - u_\alpha\|_{L^\infty(U)}, & p = \infty \end{cases}$$

$$\to 0. \quad \#$$

Thm: If $u \in W^{k,p}(U)$ and $|\alpha|+|\beta| \le k$, then

$$D^\alpha(D^\beta u) = D^{\alpha+\beta} u = D^\beta(D^\alpha u).$$

Pf: For $\varphi \in C_c^\infty(U)$,

$$\int_U D^\beta u \, D^\alpha \varphi = (-1)^{|\beta|} \int_U u \, \overbrace{D^\beta D^\alpha \varphi}^{D^{\alpha+\beta}\varphi}$$

since

$$\int_U u D^\alpha \varphi = (-1)^{|\alpha|} \int_U \varphi D^\alpha u$$

$$= (-1)^{|\beta|} \int_U u \, D^{\alpha+\beta} \varphi$$

$D^\alpha \varphi \in C_c^\infty(U)$

$$\int_U u_j D^\alpha \varphi = (-1)^{|\alpha|} \int_U \varphi D^\alpha u_j \qquad \forall \varphi \in C_c^\infty(U)$$

$$\Big\downarrow ? \qquad\qquad \Big\downarrow ?$$

$$\int_U u D^\alpha \varphi \overset{?}{=} (-1)^{|\alpha|} \int_U \varphi u_\alpha$$

$$\|u_j - u\|_{L^p(U)} \to 0, \quad \|D^\alpha u_j - u_\alpha\|_{L^p(U)} \longrightarrow 0$$

If $\frac{1}{p}+\frac{1}{q}=1$, then

<u>Hölder inequality:</u> $$\left|\int_U uv\right| \leq \|u\|_{L^p(U)} \|v\|_{L^q(U)}$$

$\forall u \in L^p(U),\ v \in L^q(U).$